

HR Recruitment AI Agent

Predicting Candidate Job-Change Likelihood from Training Program Data

Project ITI — Job Change Prediction Project | August 2026

Prepared for: Eng/Raheeq Ayman and Information Technology Institute

1. Problem Statement

Companies that fund technical training programs, such as data science bootcamps, face a recurring and costly problem: many candidates complete the training and then take a job elsewhere instead of joining or remaining with the sponsoring company. Because training is expensive in both time and resources, this turnover reduces the return on the company's investment and makes it difficult to plan recruitment and course capacity effectively.

This project defines the predictive core of an HR recruitment AI agent designed to address that problem. Rather than manually reviewing every candidate's profile, the agent consumes each candidate's demographic, educational, and employment-history data and outputs a probability that the candidate is actively looking for a new job. Recruiters can then use this score to prioritize retention outreach toward high-risk candidates and focus limited resources on the trainees most likely to leave, while spending less effort on those who are likely to stay. The end output of the agent is a ranked shortlist of candidates ordered by their estimated likelihood of seeking a job change, which becomes an actionable recommendation feed for the HR team.

2. Dataset Description

The project uses the publicly available “HR Analytics: Job Change of Data Scientists” dataset, published on Kaggle by arashnic (<https://www.kaggle.com/datasets/arashnic/hr-analytics-job-change-of-data-scientists>). The dataset originates from a company active in the Big Data and Data Science space that runs training courses for prospective hires. It was created to help the company distinguish candidates who genuinely intend to work for the company after training from those who are using the training as a stepping stone toward a job elsewhere — information that feeds directly into training investment, hiring, and course-planning decisions.

The training set contains 19,158 candidate records across 14 columns, including the binary target label (1 = looking for a job change, 0 = not looking); a separate, unlabeled test set of held-out candidates is provided for scoring. The data is a mix of numerical and categorical fields, several with high cardinality or missing values, and the target class is imbalanced — roughly a quarter of candidates are labeled as job-seeking. This combination of imbalance, mixed data types, and missing data made the dataset representative of real-world HR data and a good fit for building the recruitment agent's prediction core.

Key Features

Column	Description
enrollee_id	Unique candidate identifier
city / city_development_index	Candidate's city and its development index
gender	Candidate gender
relevent_experience	Whether the candidate has relevant work experience
enrolled_university	Current university enrollment status

Column	Description
education_level	Highest level of education completed
major_discipline	Field of study
experience	Total years of work experience
company_size / company_type	Size and type of current employer
last_new_job	Years since the candidate's last job change
training_hours	Hours of training completed
target	1 = looking for a job change, 0 = not looking (train set only)

3. Methodology

3.1 Exploratory Data Analysis

Initial analysis profiled the dataset's shape, data types, missing-value distribution (visualized with missingno), and duplicate rows. Numerical features (city_development_index, training_hours) were examined for distribution shape and outliers using boxplots and the IQR method, while categorical features (gender, relevent_experience, education_level, enrolled_university) were examined for class balance. A bivariate analysis compared the target rate across categories of education_level, company_size, and relevent_experience to identify which candidate segments were most associated with job-seeking behaviour.

3.2 Preprocessing and Feature Engineering

- Missing values across both train and test sets were filled with an explicit “Unknown” category rather than dropped, preserving every candidate record.
- Outliers in training_hours were capped using the IQR method, with bounds computed only on the training set and then applied to the test set to avoid data leakage.
- Ordinal features (education_level, relevent_experience, experience, company_size, last_new_job, enrolled_university) were mapped to ordered integer scales that respect their natural ranking.
- The high-cardinality city field was converted to a frequency encoding (share of candidates from that city) rather than one-hot encoded.
- Low-cardinality nominal features (major_discipline, gender, company_type) were one-hot encoded, with train/test columns aligned so both sets share an identical feature space.

3.3 Model Development

All models were evaluated using 5-fold stratified cross-validation to preserve the target's class balance across folds. Four modeling approaches were compared: a scaled Logistic Regression with balanced class weights as an interpretable baseline; a Random Forest classifier, first with default parameters and then tuned via RandomizedSearchCV; an XGBoost classifier, similarly tuned via RandomizedSearchCV and configured with a scale_pos_weight term to counter class imbalance; and a soft-voting ensemble combining the tuned Random Forest, tuned XGBoost, and Logistic Regression.

3.4 Evaluation Strategy

ROC-AUC was used as the primary metric because it is threshold-independent and robust to class imbalance. Because the recruitment agent's practical value depends on correctly identifying job-seeking candidates (the minority class), precision, recall, and F1-score for that class were tracked alongside ROC-AUC — a missed job-seeking candidate represents a lost recruiting opportunity, making recall on this class operationally important.

4. Results

The table below summarizes cross-validated performance for the top three models. The tuned Random Forest and tuned XGBoost models performed comparably and both clearly outperformed the Logistic Regression baseline; the soft-voting ensemble of all three models did not improve on the best single tuned model.

Model	ROC-AUC	Recall (Class 1)	Precision (Class 1)	F1 (Class 1)
Tuned XGBoost	0.803	0.735	0.559	0.635
Tuned Random Forest	0.801	0.719	0.562	0.631
Logistic Regression	0.770	0.709	0.486	0.577

Key Predictors

Feature importance from the tuned Random Forest model shows that a candidate's city development index and current company size are by far the strongest predictors of job-seeking behaviour, together accounting for more than half of total feature importance, followed by city frequency and years of experience. Personal attributes such as gender and major discipline contributed comparatively little to the model's predictions.

Feature	Importance
city_development_index	0.327
company_size	0.228
city_freq	0.121
experience	0.082
training_hours	0.075

Applying the tuned model to the held-out candidate set produces a ranked shortlist of the individuals most likely to be looking for a new job, each with an associated probability score — the direct output the recruitment agent would use to prioritize outreach.

5. Conclusion

This project built and compared four modeling approaches for predicting whether a candidate completing a data science training program is likely to be looking for a new job. The tuned XGBoost and tuned Random Forest models both reached approximately 0.80 ROC-AUC, a meaningful improvement over the Logistic Regression baseline, and produce probability scores suitable for ranking candidates by recruitment priority — forming the predictive core of an HR recruitment AI agent.

The results also have practical limits worth noting for deployment: precision on the job-seeking class remains moderate (around 0.56), meaning any outreach list generated from this model will still include a meaningful share of candidates who are not actually job-seeking, and the dataset does not include potentially informative signals such as compensation or role satisfaction. Future work could improve the agent by applying resampling techniques such as SMOTE to further address class imbalance, adding SHAP-based explanations so recruiters can see why a candidate was flagged, testing gradient-boosting alternatives such as LightGBM or a neural network, and consolidating preprocessing and modeling into a single deployable pipeline.